Measuring Counterfactual Fidelity in Machine Unlearning

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Working preprint. This manuscript reports infrastructure and exploratory pilot observations only. No confirmatory or successful-unlearning claim has been established.

Abstract

Machine-unlearning evaluations often compare an approximate intervention with a single retrained model, despite stochastic training producing a population of counterfactual models. We present an evidence-traceable framework that first tests whether a deletion request is empirically identifiable relative to exact-retraining variability and then assesses functional, privacy, representation, utility, and computational fidelity. A Fashion-MNIST pilot validates immutable artifacts and multi-seed reference estimation. Retrospective analysis finds that its one-percent IID deletion is not separable under the two evaluated output metrics, illustrating why apparent proximity to retraining need not constitute evidence of removal. This working preprint specifies the stronger prospective assay and archive contract; confirmatory method comparisons remain pending.

1 Introduction

Approximate machine unlearning is intended to remove selected training influence without the cost of complete retraining. The relevant counterfactual is a model trained on the retained data, but stochastic optimization produces a distribution of such models. This study asks whether approximate methods fall within that natural distribution across behavioral, distributional, calibration, privacy, and representation measurements. Before asking whether an approximate model matches that distribution, we ask whether the original model is distinguishable from it. If not, the deletion assay cannot support a positive or negative forgetting claim. This manuscript reports a bounded pilot observation and a prospective protocol. It makes no novelty or successful-unlearning claim.

2 Related Work

Certified removal formalizes distributional indistinguishability from retraining [2]. Partitioned training reduces deletion cost [1], while stability-based deletion analyzes deletion bounds for convex objectives [4]. Deep unlearning benchmarks compare utility and forgetting objectives [3]. Membership inference is an important attack family [6], but attack success is not itself proof of counterfactual equivalence. Generalization-oriented formulations clarify that recognizing a deleted IID example need not imply memory of that example [5]. Auditable definitions further motivate recording the algorithm, data lineage, and exclusion evidence rather than inferring deletion from outputs alone [7]. Fashion-MNIST is cited as a dataset rather than treated as a software dependency [8]. The complete review protocol is maintained in the repository.

3 Problem Formulation

For a randomized training algorithm T, deletion set $F \subset D$, and unlearning operator U, the target is the distribution $T(D \setminus F)$ rather than a single parameter vector. Counterfactual fidelity is treated as a multidimensional relation between $U(T(D), F)$ and independent draws from that distribution.

4 Counterfactual Evaluation Framework

Exact/exact comparisons define an empirical null envelope, but pairwise distances are not treated as independent observations because each trained model recurs in several pairs. Original/exact distance is compared with this envelope before any approximate-method decision. Approximate/exact comparisons are evaluated across prediction, probability, calibration, privacy, representation, local behavior, corruption response, and cost. Composite pass/fail rules and equivalence margins will be frozen before confirmatory experiments. The decision states are assay-insensitive (U0), functional failure (U1), functional-only equivalence (U2), strong empirical equivalence (U3), and inconclusive (U4).

5 Experimental Design

The completed P001 pilot uses a deterministic 9,000-example Fashion-MNIST training split, a small CNN, one frozen IID deletion request containing 90 examples, one original-model seed, and 15 exact-retraining seeds. Phase 3 adds a reference-only multi-original-seed assay over increasing deletion fractions before any approximate-method comparison. The confirmatory C001 design has not been frozen and may not begin until the separability threshold, equivalence margins, multiplicity family, manifests, and OSF preregistration are locked.

6 Pilot Results

P001 completed its artifact gate and its frozen $k = 10 \to 15$ diagnostics changed by 4.48% for mean Jensen–Shannon divergence and 3.14% for prediction disagreement. These are seed-count stability diagnostics, not equivalence margins.

A retrospective separability analysis compared the original model with the exact ensemble. The separability ratios were 0.442 for JS divergence and 0.577 for prediction disagreement. Both were below their exploratory power-derived thresholds, yielding U0 (assay-insensitive). Consequently, the pilot does not determine whether any approximate method forgot the IID-deleted examples. No confirmatory method result is reported.

7 Analysis

The pilot convergence analysis bootstraps independently trained exact replicas rather than assigning an independent sample size of $\binom{15}{2}$. Phase 3 power simulation likewise uses one summary per exact-model seed. P001 contains only one original-model seed, so its retrospective U0 classification motivates but cannot freeze the prospective assay threshold. All numerical values in the preceding section are generated from hash-bound JSON artifacts by the paper build.

8 Discussion

The pilot exposes an identifiability problem rather than a method ranking. Ordinary IID deletions may leave the trained function almost unchanged relative to ordinary retraining variation. In such a regime, approximate/exact proximity is compatible both with genuine removal and with doing essentially nothing. Stronger class-conditioned, influential, and controlled-memorization regimes are therefore required. This interpretation remains restricted to assay design and does not establish the behavior of any unlearning method.

9 Limitations and Ethics

The current evidence covers one reduced Fashion-MNIST split, a small CNN, one deletion manifest, and one original-model seed in P001. The multi-original assay remains a pilot. Approximate unlearning does not by

itself establish legal compliance or demonstrate the absence of private information. Membership attacks are incomplete adversaries, and representation or parameter similarity does not uniquely determine functional influence.

10 Broader and Ethical Considerations

Empirical similarity to a finite retraining population is not proof that personal data has been erased, nor is it a legal compliance certificate. Attack results depend on the adversary and data composition. Releases must avoid exposing deleted or sensitive records.

11 Conclusion

P001 demonstrates a reproducible pipeline and motivates reference separability as a precondition for empirical unlearning claims. Its one-percent IID deletion is assay-insensitive under the evaluated output metrics. Whether approximate methods attain functional, privacy, representation, and utility equivalence remains an open empirical question for the prospectively registered study.

A Reproducibility Appendix

P001 is identified by 'fashion-cnn-random-01pct-schema2-rocm-pilot-v1'; its exact seeds are 2001–2015 and its deletion manifest is archived under 'experiments/P001'. Original training took 9.59 seconds and all exact retrainings took 17.46 seconds on the recorded RX 7800 XT/ROCm environment. These device measurements are not general performance estimates. The repository stores complete configurations, failures, environment metadata, prediction/checkpoint hashes, and figure source provenance.

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